

UNIT-5

Transport Layer & Application Layer

2024

Transport Layer → The Transport layer is the 4th layer of the OSI Model.

It provides end-to-end communication between source and destination processes.

It ensures that data is delivered reliably, correctly and in order.

• The Transport layer breaks large data into smaller segments and sends them to the destination.

Design Issues of Transport Layer

These are the problems handled by the transport layer.

- a) Addressing → Identifies the sending and receiving processes using port numbers.
- b) Connection Establishment → Creates a connection ~~between~~ before data transfer.

- c) Flow Control - Controls data transmission speed.
- d) Error Control → Detects and correct errors.
- e) Multiplexing → Allows multiple applications to use the network simultaneously.

UDP (User Datagram Protocol)

UDP (User Datagram Protocol) is a connectionless transport layer protocol that sends data without establishing a connection between sender and receiver.

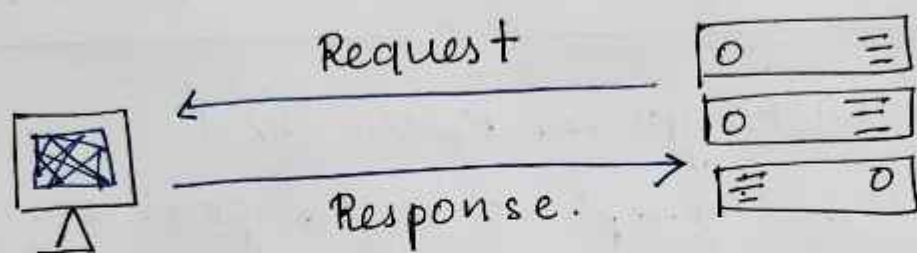
It is fast but does not guarantee reliable delivery of data.

Features of UDP

1. Connectionless Protocol → No connection setup before sending data
2. Fast Transmission → Data is sent quickly.
3. No error Recovery → Lost packets are not retransmitted.
4. Low Overhead → Small header size (8 byte).
5. Unreliable Communication → No guarantee of delivery.

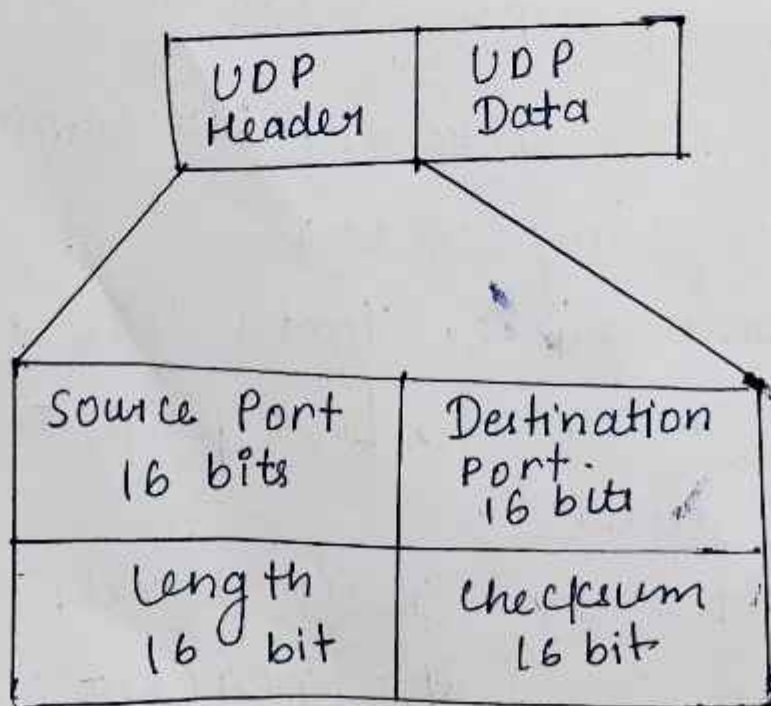
UDP Working

- 1→ Sender creates a UDP packet (Datagram)
- 2→ Packet is sent directly to the receiver
- 3→ Receiver accepts the packet if it arrives.
- 4→ No acknowledgment is sent back.



UDP : Header format 2024, 2025

UDP Header is very simple and consists of 4 fields, each of 16 bits (2 bytes)



Fields of UDP Header

1. Source Port (16 bits)

- Identifies the sending application.

Example → Port 80, Port 53.

2. Destination (16 bits)

- Identifies the receiving application.
- Helps deliver data to the correct process.

3. Length (16 bits)

- Specifies the total length of the UDP header and data.
- Minimum length = 8 bytes

4. Checksum (16 bits)

- Used for error detection.
- Checks whether data is corrupted during transmission

UDP Per-Segment Checksum

Per-Segment Checksum is an error-detection mechanism used in UDP (User Datagram Protocol) to check whether data is corrupted during transmission

Working-

1. Sender calculates a checksum value using:-
 - UDP Header
 - Data (Payload)
 - Pseudo Header (contains source IP, destination IP, protocol, and length)
2. The checksum is placed in the UDP header
3. Receiver performs the same calculation:
 - If the calculated checksum matches the received checksum → Data is accepted
 - If it does not match → segment is discarded

Why it is needed

Detects transmission errors

Ensures Data Integrity

Helps identify corrupted packets

Carrying Unicast / Multicast real time traffic

Real time applications such as video conferencing, online gaming, and live streaming require data to be delivered quickly with minimal delay.

UDP is commonly used because it has low overhead.

1. Unicast Real - Time Traffic

Unicast communication involves one sender and one receiver.



Characteristics

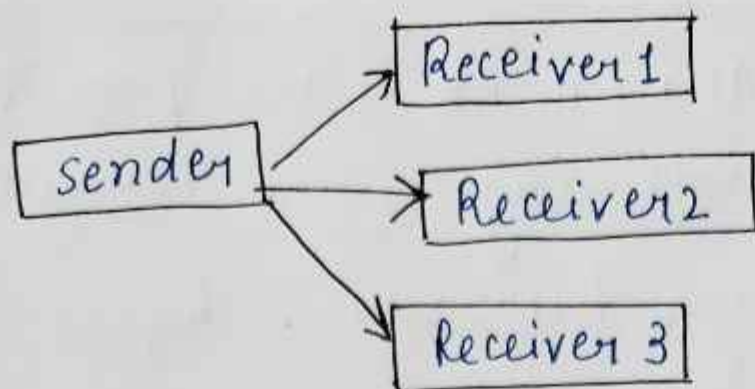
- One-to one communication
- Separate packet sent to each receiver
- Higher bandwidth consumption when many receivers exist.

Example

- Voice call between two people
- Video call
- web browsing

② Multicast Real-Time Traffic

Multicast communication involves one sender and multiple receivers belonging to a multicast group



Characteristics

- One to many communication
- Sender transmits only one copy
- Network duplicates packets where necessary.

Examples.

- live video streaming
- Online classes
- IPTV
- stock market updates

TCP [Imp]

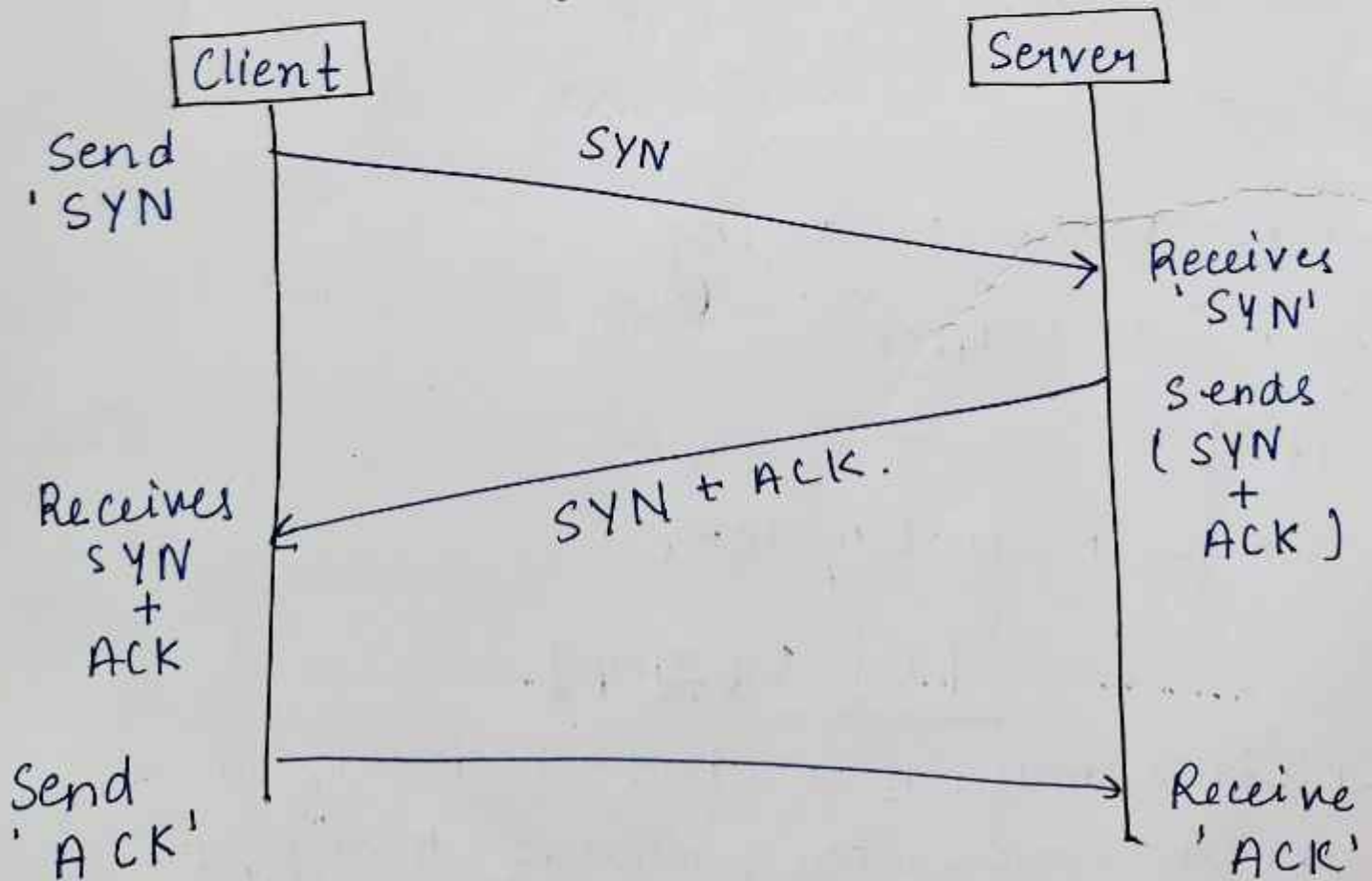
TCP (Transmission Control Protocol) is a connection-oriented, reliable transport layer protocol that ensures data is delivered accurately and in the correct order between sender and receiver.

TCP Connection Management

TCP connection Management is the process of establishing, maintaining, and terminating a connection between two devices before and after data transmission.

① Connection Establishment (Three Way Handshake)

Before data transfer, TCP creates a connection using the Three-way Handshake.



Steps.

- ① SYN → Client sends a SYN (synchronize) packet
 - Requests connection establishment.
- ② SYN + ACK : Server accepts the request.
 - Sends SYN + ACK packet.
- ③ ACK :- Client sends ACK
 - Connection is successfully established.

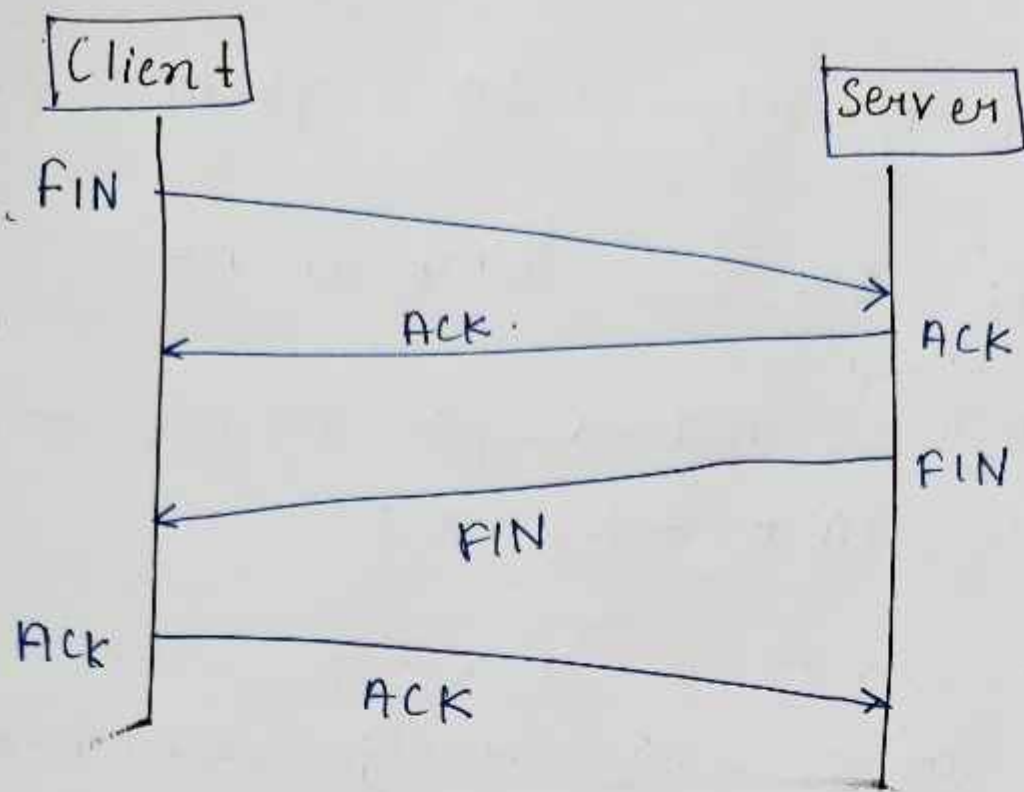
② Data Transfer

★ After connection establishment.

- Data is divided into segments
- Sequence numbers are assigned.
- Receiver sends acknowledgement.
- Lost segments are retransmitted.

③ Connection Termination (Four-Way Handshake)

→ When communication ends, TCP closes the connection



Steps

- 1 Client sends FIN
- 2 Server sends ~~ack~~ ACK
- 3 Server sends FIN
- 4 Client sends ACK.

★ Connection is terminated

Reliability of Data Transfer

Reliability of Data Transfer means ensuring that data sent by the sender reaches the receiver accurately, completely and

in the correct order. without loss or duplication.

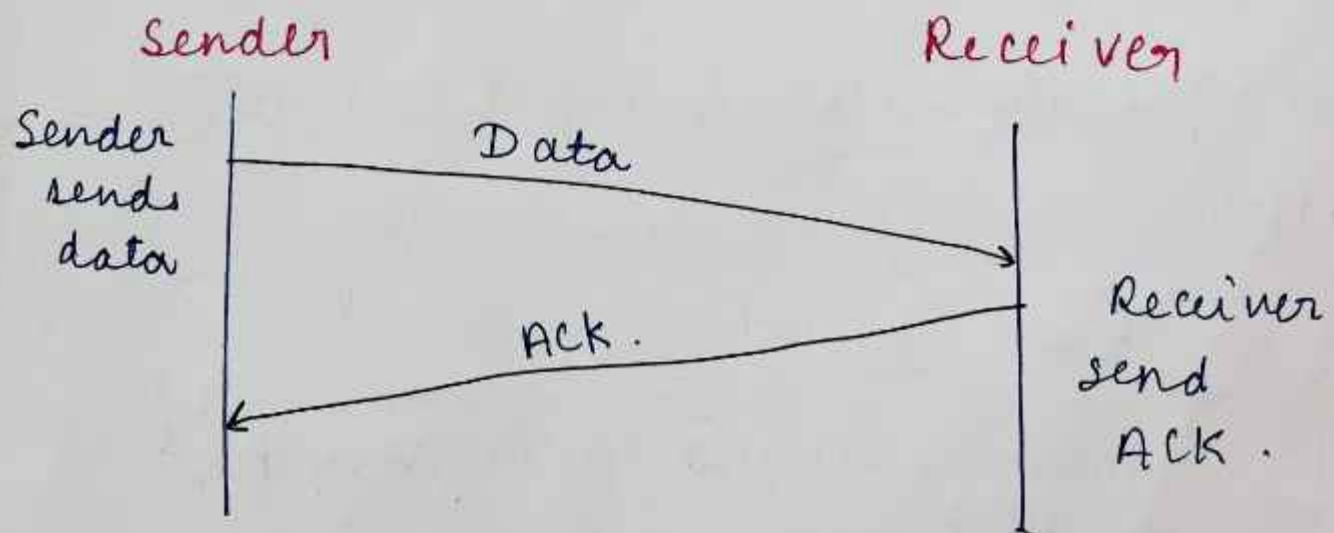
How TCP Provides Reliable Data Transfer

1. Sequence Numbers

- Each TCP segment is assigned a unique sequence number.
- Helps the receiver arrange data in the correct order

2. Acknowledgements (ACK)

- Receiver sends an ACK after receiving data successfully.
- Confirms successful delivery.



3. Retransmission

- If an ACK is not received within a specified time, TCP retransmits the segment.
- Ensures lost data reaches the receiver.

4. Checksum

- Used to detect errors in transmitted data.
- Corrupted segments are discarded and retransmitted.

TCP flow Control

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TCP flow control is a mechanism that prevents the sender from sending data faster than the receiver can process it. It ensures that the receiver's buffer does not overflow.

Need for Flow Control

- Sender may transmit data very fast
- Receiver may process data slowly.

- Without flow control, data can be lost due to buffer overflow.

Sliding Window Mechanism

TCP uses a Sliding Window Protocol for flow control.

- The sender can send multiple segments before receiving an ACK.
- The receiver advertises the available buffer size (window size).
- As ACKs arrive the window "slides" forward.

Advantages.

- Prevents receiver buffer overflow
- Improves transmission efficiency
- Ensures reliable communication.

TCP Congestion Control 2024 May 2023

TCP ~~from~~ congestion control is a mechanism used to prevent excessive traffic in the network. It controls the rate at which the sender transmits data to avoid network congestion and packet loss.

Need for congestion control.

Too much data in the network can cause congestion.

Congestion leads to

Packet loss

Increased loss

Reduced network performance.

TCP adjust the network rate according to network conditions.

Technique of TCP congestion control.

1. Slow Start →

TCP begins with a small congestion window (cwnd).

- Window size increases exponentially after each ACK.

$1 \rightarrow 2 \rightarrow 4 \rightarrow 8 \rightarrow 16$.

Purpose $\rightarrow$ Quickly find available network capacity.

②. Congestion Avoidance

- After reaching a threshold, window size grows slowly.
- Increase is linear rather than exponential.

$16 \rightarrow 17 \rightarrow 18 \rightarrow 19$.

Purpose $\rightarrow$ Avoid sudden congestion

③ Fast Retransmit

- If the sender receives 3 duplicate ACKs it assumes a packet is lost
- Lost packet is retransmitted immediately

Benefit $\rightarrow$ No need to wait for timeout.

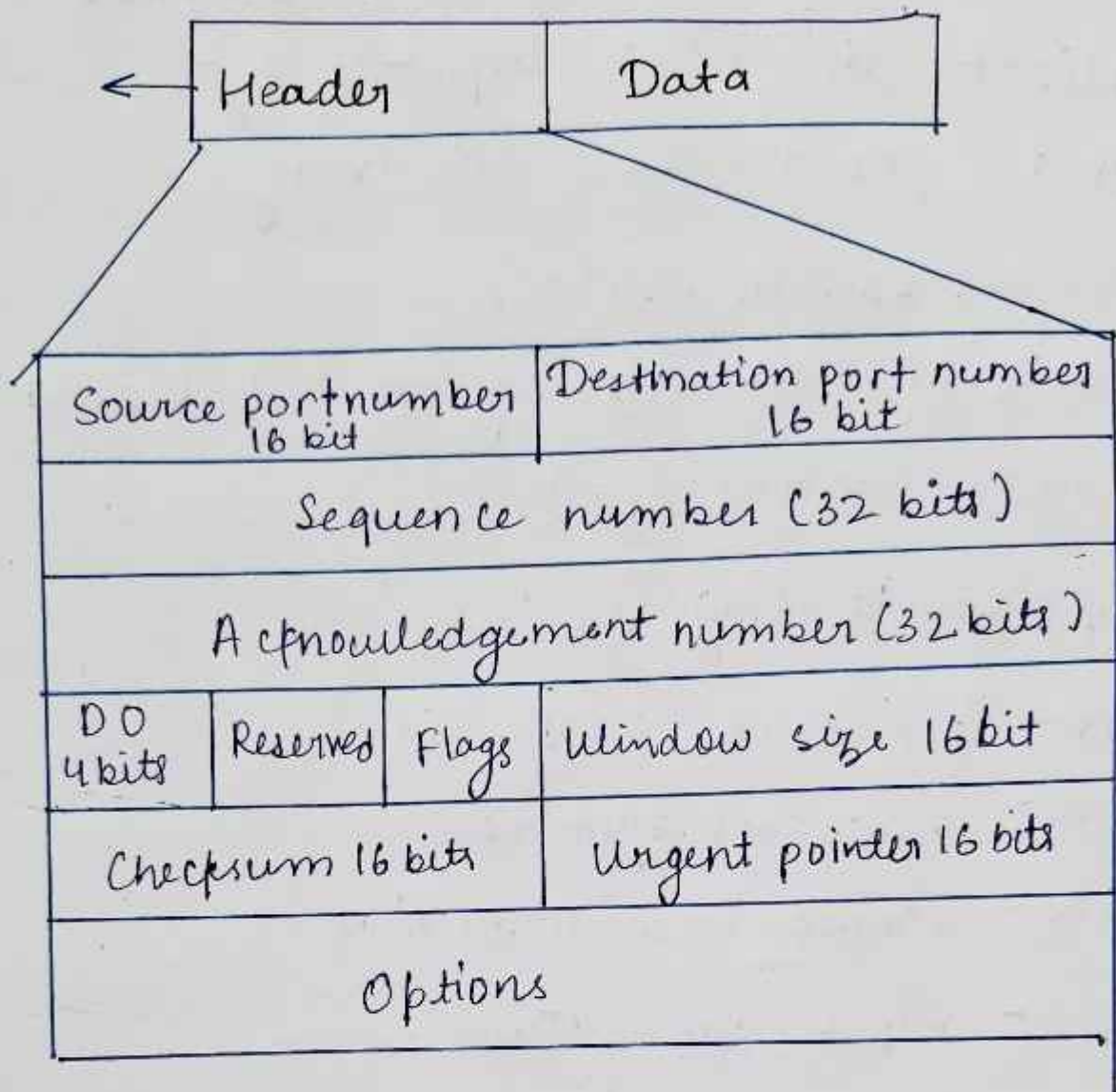
④ Fast Recovery.

- After fast Retransmit, TCP reduces the congestion window.
- Transmission continues without returning to the initial slow start phase

Benefit → Faster recovery from packet loss.

TCP Header Format 2025

The TCP Header contains control information required for reliable data transmission between sender and receiver.



Fields of TCP Header.

- 1] Source Port (16 bits) → Identifies the sender's application.
- 2] Destination Port (16 bits) → Identifies the receiver's application.
- 3] Sequence Number (32 bits) → Specifies the position of data in the stream.
 - Helps maintain correct order.

4. Acknowledgment Number (32 bits)
- Indicates the next expected byte.
 - Used for reliable delivery.

5. Header Length (4 bits)
- Specifies the size of the TCP header

6. Flags (Control Bits)

Important flags are:-

- SYN - Establish connection
- ACK - Acknowledgment
- FIN - Terminate connection
- RST - Reset connection.
- PSH - Push data immediately
- URG - Urgent data.

7. Window Size (16 bits)

- Used for flow control
- Specifies how much data can be received

8. Checksum (16 bits)

- Detects transmission errors.

9. Urgent Pointer (16 bits)

- Points to urgent data.

10. Options (Variable length)

- Additional TCP features and parameters.

Application Layer

The Application layer is the top most layer of the TCP/IP model and OSI model.

It provides network services directly to end users and enables communication between applications over a network.

Functions of Application Layer.

1. Provides user interface to network services.
2. Supports file transfer.
3. Enables email communication.
4. Supports web browsing.
5. Provides remote login services.
6. Facilitates network resource sharing.

Common Application layer Protocols .

- HTTP
- FTP
- SMTP
- DNS
- SSH

WWW (World Wide Web)

The World Wide Web (WWW) is a collection of interconnected web pages and resources available on the internet, which can be accessed using a web browser.

WWW was invented by Tim Berners-Lee in 1989.

2020 HTTP (Hyper Text Transfer Protocol)

HTTP is an Application Layer Protocol used for communication between a web browser (client) and a web server.

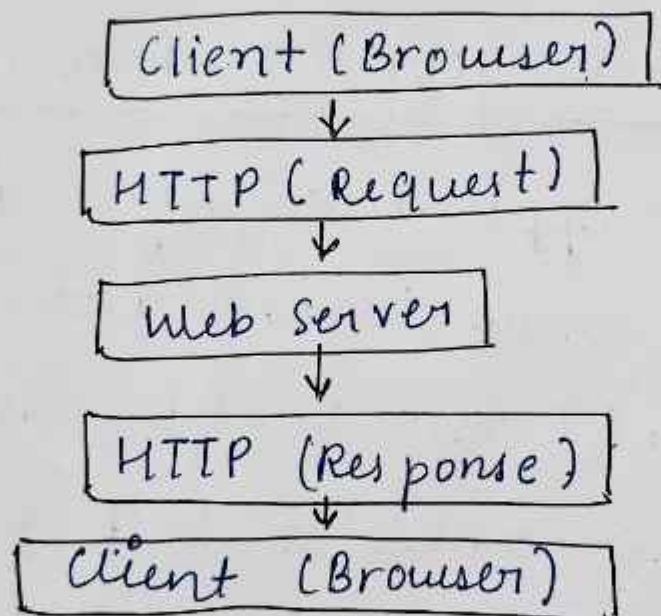
It is used to transfer web pages and other resources on the World Wide Web (WWW).

Features

1. Client - server architecture
2. Request - Response protocol.
3. Support text, images, audio, and video.
4. Simple and fast communication.

Working of HTTP

1. User enters a URL in the browser.
2. Browser sends an HTTP request to the server.
3. Server processes the request
4. Server sends an HTTP response
5. Browser displays the web pages.



HTTP Request Methods.

1. GET → Retrieves data from the server
2. POST → Sends data to the server
3. PUT → Updates existing data
4. DELETE → Removes data from the server.

HTTP Response Status Codes.

code	Meaning
200	OK (success)
404	Page not found
500	Internal Server Error
301	Moved Permanently.

FTP (File Transfer Protocol)

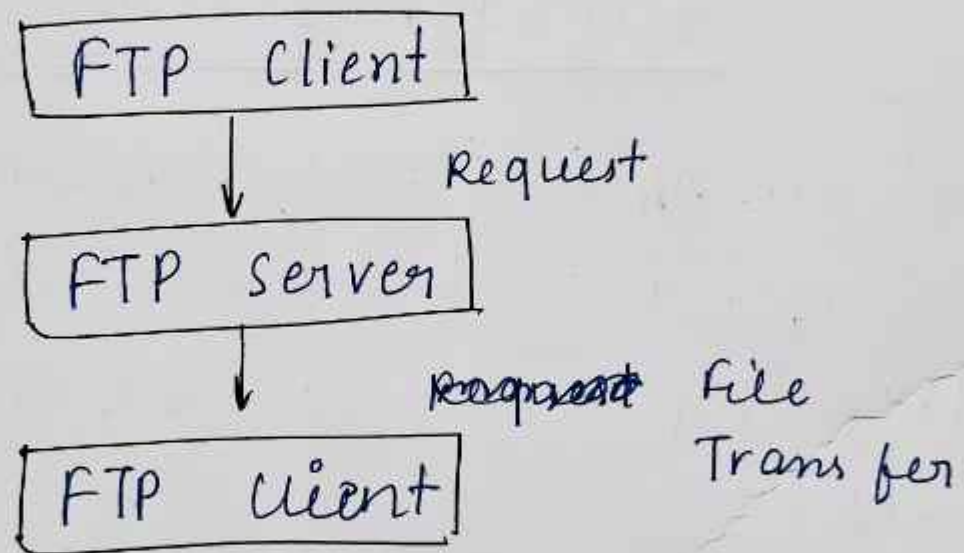
FTP is an Application layer Protocol used to transfer files between a client and a server over a network such as the internet.

Features of FTP

- 1 File upload and Download
- 2 Client-server architecture
- 3 Reliable data transfer using TCP
- 4 Supports authentication (username and password)
- 5 Allow remote file access.

Working of FTP

1. Client connects to the FTP server
2. User logs in with username & password
3. Client requests file upload / Download.
4. Server transfers the requested file
5. Connection is closed after transfer



FTP Connections.

FTP uses two TCP connections

① Control Connection - Used for command and responses.

- Port Number = 21

② Data Connection

- Used for actual file transfer
- Port Number : 20.

Applications. ① Uploading website file to a server
② Downloading software
③ Sharing large file
④ Backup and data transfer.

SSH (Secure Shell).

SSH is an Application Layer Protocol used for secure remote login and communication between two computers over a network.

Features of SSH

1. Secure remote access
2. Encrypted communication
3. Authentication using username / password
4. Protect data from unauthorized access.
5. Support secure file transfer.

Working.

1. Client requests connection to the SSH server
2. Server authenticates the user
3. An encrypted communication channel is established.

User can execute commands remotely and securely.

SSH Port Number.

- Port 22 (Default Port).

Email (Electronic Mail)

E-mail is an application layer service used to send and receive digital message over the internet.

Component of E-mail

① → User Agent (Mail client)

Software used to compose, send and receive emails.

Examples → Gmail, Microsoft Outlook.

2. Mail server - Stores and forwards email messages.
3. mailbox - Storage area where received emails are kept.

E-mail Protocols.

1. SMTP
2. MIME
3. IMAP

① SMTP (Simple Mail Transfer Protocol)

SMTP is an Application Layer protocol used for sending and forwarding emails over the internet.

Functions of SMTP

- Sends emails from sender to mail server.
- Transfers emails between mail server.
- Delivers outgoing email messages

Working

- 1) User composes an email
- 2) Email client sends the message to the sender's mail server using SMTP.

- 3) The sender's mail server forwards the email to the receiver's mail server
- 4) The receiver accesses the email using POP3 or IMAP.

Features of SMTP

- 1 Simple and reliable email transfer
- 2 Works on client-server architecture
- 3 Used only for sending email
- 4 Supports text and attachments.

SMTP Commands.

- HELO / EHLO → Identify client
- MAIL / FROM → Specify sender
- RCPT TO → Specify recipient
- DATA → Send message content
- QUIT → End session.

★ SMTP used only for sending email

②

MIME

MIME stands for Multipurpose Internet Mail Extensions

It is a standard used in the Application Layer (especially in email system like SMTP)

MIME helps to send different types of data (not just plain text) over the internet

Why MIME is needed

Originally, email systems could only send simple text (ASCII characters).

Problem → cannot send images, videos, and files (PDF, DOC etc).

So, MIME solve this problem by allowing
→ Images, Audio, Video, Documents.

MIME is mainly used with

- Email protocols (like SMTP)
- Also used in HTTP (web browsers) to identify file types.

③ IMAP

IMAP stands for Internet message Access Protocol.

It is an Application layer protocol used to receive emails from a mail server.

IMAP allows you to read emails directly from the server without downloading them permanently.

This means :-

- You can access the same email from mobile, laptop, tablet.
- Email stay stored on the mail server.

How IMAP works.

- ① You open your email app (like Gmail, Outlook)
- ② IMAP connects to the main server

3. Emails are shown but not fully downloaded.

4. Any changes (read, delete, move) are saved on the server

Features

- 1 Email stored on server
- 2 Access from multiple devices
- 3 Synchronization (same view everywhere)
- 4 Supports folders (Inbox, Sent, Drafts)
- 5 Allows partial download (only header or part).

IMAP default port numbers.

- Port 143 → Non-secure
- Port 993 → Secure (IMAPS)

DNS

DNS stands for Domain Name system.

It is an Application Layer Protocol used to convert website names into IP addresses.

Example → Humans remember names like

- google.com
- Youtube.com

But computers understand numbers like

→ 142.250.183.14

DNS converts name → IP address.

Working of DNS

- 1 You type www.google.com in browser
- 2 Request goes to DNS server
- 3 DNS server finds the IP address
- 4 Returns IP to browser
- 5 Browser connects to that IP → website opens.

Types of DNS.

- 1 Root DNS server
- 2 TLD server (.com, .org, .in)
- 3 Authoritative DNS server

★ These work together to find correct IP address

DNS uses port Number.

Port 53 (UDP/TCP)

SNMP

2024

SNMP stands for Simple Network Management protocol. It is a standard protocol used to monitor and manage network devices such as routers, switches, servers, printers, firewalls, and IoT devices.

Main components.

- SNMP Manager (NMS): - The monitoring system (e.g. Zabbix, SolarWinds, Nagios).
- SNMP Agent: Software running on the managed device that collects information.
- MIB (Management Information Base): A database of objects / OIDs that can be queried.
- OID (Object Identifier): - Unique identifier for a specific metric or parameter.

How it Works.

1. The SNMP manager sends a request to device.

2. The SNMP agent responds with the request information.
3. The agent can also send trap or inform messages to the manager when event occur.

Common Operations.

- GET → Retrieve a value.
- GETNEXT - Retrieve the next OID (Object Identifier) value.
- GETBULK - Retrieve multiple values.
- SET → Change a configuration value.

SNMP Versions.

Version	Security.
SNMPV1 →	Basic. community strings only
SNMPV2C →	Improved performance, still community - string based
SNMPV3 →	Authentication and encryption (recommended).

Common Ports.

- UDP 161 → SNMP queries.
 - UDP 162 → SNMP traps.
-